

Experiment – 08 | Views in PostgreSQL

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Class Task

Class Task: Views in PostgreSQL

Medium Level – Create a complex view from customer_master, product_catalog and sales_orders, exposing only Order ID, Order Date, Customer Name, Product Name and Final Amount. Gross Amount = Unit Price × Quantity; Discount Amount = Gross Amount × Discount Percentage / 100; Final Amount = Gross Amount – Discount Amount.

1. Create source tables and insert the data given in the PPT.

```
CREATE TABLE customer_master (  
    customer_id VARCHAR(10) PRIMARY KEY,  
    full_name VARCHAR(100),  
    city VARCHAR(50)  
);  
  
CREATE TABLE product_catalog (  
    product_id VARCHAR(10) PRIMARY KEY,  
    product_name VARCHAR(100),  
    unit_price NUMERIC(12,2)  
);  
  
CREATE TABLE sales_orders (  
    order_id INT PRIMARY KEY,  
    product_id VARCHAR(10) REFERENCES product_catalog(product_id),  
    quantity INT,  
    customer_id VARCHAR(10) REFERENCES customer_master(customer_id),  
    discount_percent NUMERIC(5,2),  
    order_date DATE  
);  
  
INSERT INTO customer_master VALUES  
( 'C1', 'Amit Sharma', 'Delhi'),  
( 'C2', 'Priya Verma', 'Mumbai'),  
( 'C3', 'Ravi Kumar', 'Bangalore'),  
( 'C4', 'Neha Singh', 'Kolkata'),  
( 'C5', 'Arjun Mehta', 'Hyderabad');  
  
INSERT INTO product_catalog VALUES  
( 'P1', 'Smartphone X100', 25000),  
( 'P2', 'Laptop Pro 15', 65000),  
( 'P3', 'Wireless Earbuds', 5000),  
( 'P4', 'Smartwatch Fit', 30000),  
( 'P5', 'Gaming Console', 45000);  
  
INSERT INTO sales_orders VALUES  
(1, 'P1', 2, 'C1', 5, '2025-09-01'),  
(2, 'P2', 1, 'C2', 10, '2025-09-02'),  
(3, 'P3', 3, 'C3', 0, '2025-09-03'),  
(4, 'P4', 1, 'C4', 8, '2025-09-04'),
```

```
(5, 'P5', 1, 'C5', 12, '2025-09-05'),
(6, 'P1', 1, 'C2', 5, '2025-09-06'),
(7, 'P3', 2, 'C1', 0, '2025-09-07');
```

2. Create the View

```
CREATE OR REPLACE VIEW order_report AS
SELECT
    so.order_id,
    so.order_date,
    cm.full_name AS customer_name,
    pc.product_name,
    ROUND(
        (pc.unit_price * so.quantity)
        - ((pc.unit_price * so.quantity) * so.discount_percent / 100),
        2
    ) AS final_amount
FROM sales_orders AS so
JOIN customer_master AS cm
    ON so.customer_id = cm.customer_id
JOIN product_catalog AS pc
    ON so.product_id = pc.product_id;
```

3. Display the View

```
SELECT * FROM order_report ORDER BY order_id;
```

Expected output

order_id	order_date	customer_name	product_name	final_amount
1	2025-09-01	Amit Sharma	Smartphone X100	47500.00
2	2025-09-02	Priya Verma	Laptop Pro 15	58500.00
3	2025-09-03	Ravi Kumar	Wireless Earbuds	15000.00
4	2025-09-04	Neha Singh	Smartwatch Fit	27600.00
5	2025-09-05	Arjun Mehta	Gaming Console	39600.00
6	2025-09-06	Priya Verma	Smartphone X100	23750.00
7	2025-09-07	Amit Sharma	Wireless Earbuds	10000.00

4. Create the external client and restrict access

```
CREATE USER client_user WITH PASSWORD 'Client@123';

REVOKE ALL ON customer_master FROM client_user;
REVOKE ALL ON product_catalog FROM client_user;
REVOKE ALL ON sales_orders FROM client_user;

GRANT SELECT ON order_report TO client_user;

REVOKE ALL ON customer_master FROM PUBLIC;
REVOKE ALL ON product_catalog FROM PUBLIC;
REVOKE ALL ON sales_orders FROM PUBLIC;
```

The client can run SELECT on the view but has no direct access to the three source tables and cannot modify the view data.

Verification

```
SET ROLE client_user;  
SELECT * FROM order_report;  
RESET ROLE;
```